三、玻-爱分布与费-狄分布

(玻-爱) 同上，建模：粒子不可分辨 (全同球入袋) (隔板法) $|00|0\ 0|0|0$

$$\Omega_{B,E}=\prod_i\frac{(w_i+a_i-1)!}{(a_i)!\,(w_i-1)!}=\prod_i C_{w_i+a_i-1}^{w_i-1}$$

$\hookrightarrow\ \delta\ln\Omega=0\ \rightarrow\ \delta\big((w_i+a_i-1)\ln(w_i+a_i-1)-a_i\ln a_i-(w_i-1)\ln w_i\big)=0$

$w_i, a_i\gg1$，1可去

$\delta N=\sum\delta a_i=0$

$\delta E=\sum\varepsilon_i\delta a_i=0$

$\rightarrow\ (\ln(w_i+a_i)-\ln a_i)\,\delta a_i=0$

$\ln\left(\frac{w_i+a_i}{a_i}\right)=\alpha+\beta\varepsilon_i\ \rightarrow\ a_i=\frac{w_i}{e^{\alpha+\beta\varepsilon_i}-1}$ 变化

取巨配分函数 $\Xi=\prod(1-e^{-\alpha-\beta\varepsilon_i})^{-w_i}$　$\left(\ln\Xi=\sum-w_i\ln(1-e^{-\alpha-\beta\varepsilon_i})\right)$

1) $N=-\frac{\partial}{\partial\alpha}\ln\Xi=\sum\frac{w_i}{e^{\alpha+\beta\varepsilon_i}-1}$

2) $U=-\frac{\partial}{\partial\beta}\ln\Xi$　　$\frac{\partial\ln\Xi}{\partial q}=\frac{\partial\varepsilon_i}{\partial q}\cdot\frac{\partial\ln\Xi}{\partial\varepsilon_i}$ $\hookrightarrow -\frac{1}{\beta}a_i$

3) $Y_\alpha=\frac{\partial U}{\partial q_\alpha}=\sum\frac{\partial\varepsilon_i}{\partial q_\alpha}\cdot a_i=-\frac{1}{\beta}\frac{\partial}{\partial q_\alpha}\ln\Xi$

3.5) $P=\frac{1}{\beta}\frac{\partial}{\partial V}\ln\Xi$

4) 同上，　$\left(dU-PdV+\frac{\alpha}{\beta}dN\right)=\frac{1}{\beta}d\left(\ln\Xi-\alpha\frac{\partial}{\partial\alpha}\ln\Xi-\beta\frac{\partial}{\partial\beta}\ln\Xi\right)$

$dQ=TdS=(dU-PdV-\mu dN)$

$$\beta=\frac{1}{kT}\ ,\quad \alpha=\frac{-\mu}{kT}$$

5) $$S=k\left(\ln\Xi_1-\alpha\frac{\partial}{\partial\alpha}\ln\Xi_1-\beta\frac{\partial}{\partial\beta}\ln\Xi_1\right)$$
$$=k\left(\ln\Xi_1+\alpha N+\beta U\right)$$
$$=\ln\Omega$$

$\ln\Omega=$

$$\ln\Omega=\sum\left((\omega_i+a_i)\ln(\omega_i+a_i)-a_i\ln a_i-\omega_i\ln\omega_i\right)$$

$$\ln\left(\frac{\omega_i+a_i}{a_i}\right)=\alpha+\beta\varepsilon_i$$

$$\ln\Xi_1=-\sum\omega_i\ln(1-e^{-\alpha-\beta\varepsilon_i})$$

$$\hookrightarrow\ \ln\Xi_1+\alpha N+\beta U=-\sum\omega_i\ln(1-e^{-\alpha-\beta\varepsilon_i})+\ln\left(\frac{\omega_i+a_i}{a_i}\right)\cdot\frac{\omega_i}{e^{\alpha+\beta\varepsilon_i}-1}$$
$$=-\sum\omega_i\ln\left(\frac{\omega_i}{\omega_i+a_i}\right)+\ln\left(\frac{\omega_i+a_i}{a_i}\right)a_i$$
$$=\ln\Omega$$

费-狄： 费米子，泡利不相容： 一球一袋

$$\Omega_{F\text{-}D}=\prod_i\frac{\omega_i!}{a_i!(\omega_i-a_i)!}=\prod_i C_{\omega_i}^{a_i}$$

$$\downarrow$$

$$\ln\Omega=\sum\omega_i\ln\omega_i-a_i\ln a_i-(\omega_i-a_i)\ln(\omega_i-a_i)$$

$$\alpha\delta N+\beta\delta E+\delta\ln\Omega=0=-\delta a_i\left(\ln\frac{\omega_i-a_i}{a_i}-\alpha-\beta\varepsilon_i\right)\ \rightarrow\ a_i=\frac{\omega_i}{e^{\alpha+\beta\varepsilon_i}+1}$$

其余关系同玻-爱， $\Xi_{F\text{-}D}=\prod_i(1+e^{-\alpha-\beta\varepsilon_i})^{\omega_i}$

四. 玻色子, 费米子

弱简并气体: e^{α} 与 $n\lambda^3$ 可取一阶小量:

$\varepsilon=\frac{1}{2m}(P_x^2+P_y^2+P_z^2) \longrightarrow$ V上 ε到ε+dε 状态数 $D(\varepsilon)d\varepsilon = g\frac{V\cdot 2\pi(2m)^{\frac{3}{2}}\varepsilon^{\frac{1}{2}}}{h^3}d\varepsilon$ (g: 自旋简并度)

±上为费子, 下为玻子

$$N=\frac{gV\cdot 2\pi(2m)^{\frac{3}{2}}}{h^3}\int_0^{\infty}\frac{\sqrt{\varepsilon}\,d\varepsilon}{e^{\alpha+\beta\varepsilon}\pm 1}$$

$$U=\frac{gV\cdot 2\pi(2m)^{\frac{3}{2}}}{h^3}\int_0^{\infty}\frac{\varepsilon^{\frac{3}{2}}d\varepsilon}{e^{\alpha+\beta\varepsilon}\pm 1}$$

$\xrightarrow{x=\beta\varepsilon}$

$$N=\frac{gV\cdot 2\pi(\frac{2m}{\beta})^{\frac{3}{2}}}{h^3}\int_0^{\infty}\frac{\sqrt{x}\,dx}{e^{\alpha+x}\pm 1}$$

$$U=\frac{gV\cdot 2\pi(\frac{2m}{\beta})^{\frac{3}{2}}}{h^3\beta}\int_0^{\infty}\frac{x^{\frac{3}{2}}dx}{e^{\alpha+x}\pm 1}$$

小量近似 $\frac{1}{e^{a+x}\pm 1}\xrightarrow[e^{-a}\ll 1]{\text{展开}} e^{-a-x}(1\mp e^{-a-x})$

$$N=g\left(\frac{2\pi mkT}{h^2}\right)^{\frac{3}{2}}Ve^{-a}\left(1\mp\frac{e^{-a}}{2^{\frac{3}{2}}}\right),\quad U=\frac{3}{2}g\left(\frac{2\pi mkT}{h^2}\right)^{\frac{3}{2}}Ve^{-a}kT\left(1\mp\frac{e^{-a}}{2^{\frac{5}{2}}}\right)$$

相除, e^{-a} 近似: $U=\frac{3}{2}NkT\left(1\pm\frac{e^{-a}}{4\sqrt{2}}\right)$

当 $e^{-a}\to 0$ 时, N用玻尔兹曼分布结果 $e^{-a}=\frac{N}{gV}\left(\frac{h^2}{2\pi mkT}\right)^{\frac{3}{2}}$

或: $U=\frac{3}{2}NkT\left(1\pm\frac{n\lambda^3}{4\sqrt{2}}\right)$ — 费子会排斥, 玻子会吸引

经典表玻分布 全同性附加内能

玻色-爱因斯坦凝聚

玻色子 考虑 $\varepsilon = \varepsilon_l - \mu$ (化学势)，$\varepsilon > 0$ (这里有ε_l最低为0(基态)，则$\mu<0$)，ε_l上的粒子数 $a_l = \frac{\omega_l}{e^{\frac{\varepsilon}{kT}}-1}$，

μ由 $n = \frac{N}{V} = \frac{1}{V}\sum \frac{\omega_l}{e^{\frac{\varepsilon}{kT}}-1}$ 确定 (数密度)

$\frac{2\pi}{h^3}(2m)^{\frac{3}{2}}\int \frac{\sqrt{\varepsilon}\,d\varepsilon}{e^{\frac{\varepsilon_l-\mu}{kT}}-1} = n$，$T\uparrow$，$\mu\uparrow$，$T=T_0, \mu\to -0$ $\to$ $\frac{2\pi}{h^3}(2m)^{\frac{3}{2}}\int_0^\infty \frac{\sqrt{\varepsilon}\,d\varepsilon}{e^{\frac{\varepsilon}{kT_0}}-1} = n$ $\to T_0 = \frac{2\pi\hbar^2}{(2.612)^{\frac{2}{3}}}\frac{(n)^{\frac{2}{3}}}{mk}$ (积分结果)

$T<T_0$，有：$\mu\to -0$，但积分中边界值不能被视为0舍去：$\varepsilon=0$ 粒子凝聚 $n_0(T)$

$$n_0(T) + \frac{2\pi}{h^3}(2m)^{\frac{3}{2}}\int_0^\infty \frac{\sqrt{\varepsilon}\,d\varepsilon}{e^{\frac{\varepsilon}{kT}}-1} = n$$

$$\hookrightarrow n_0(T) = n\left[1-\left(\frac{T}{T_0}\right)^{\frac{3}{2}}\right]$$

$n_0(T)$粒子无能量 ($\varepsilon=0$) $\to$ $U = \frac{2\pi}{h^3}(2m)^{\frac{3}{2}}\int \frac{\varepsilon^{\frac{3}{2}}\,d\varepsilon}{e^{\frac{\varepsilon}{kT}}-1} = 0.770NkT\left(\frac{T}{T_c}\right)^{\frac{3}{2}}$，$C_v = \frac{\partial U}{\partial T} = \frac{5}{2}\frac{U}{T}$